

Unit 4: Machine Learning for Data Analytics

Complete Hinglish Notes + Algorithms + Solved Examples

1. Introduction to Machine Learning

Machine Learning (ML) ek technique hai jisme computer data se patterns seekhta hai aur automatically prediction/decision leta hai.

Example:

- Netflix recommendations
 - Spam email detection
 - Face recognition
 - Weather prediction
-

Types of Machine Learning

A) Supervised Learning

Yahan model ko labeled data diya jata hai.

Labeled Data = Input + Correct Output

Example

Hours Studied	Result
2	Fail
5	Pass
8	Pass

Model seekhega ki kitne study hours par student pass/fail hota hai.

Types of Supervised Learning

1. Classification

Output categories me hota hai.

Examples:

- Spam / Not Spam
- Pass / Fail
- Disease / No Disease

2. Regression

Output numerical value hoti hai.

Examples:

- House price prediction
- Temperature prediction

B) Unsupervised Learning

Yahan data unlabeled hota hai.

Model khud hidden patterns ya groups find karta hai.

Example

Customers ko groups me divide karna:

- Students
- Professionals
- Premium users

Supervised vs Unsupervised

Feature	Supervised	Unsupervised
Labels	Present	Not Present
Goal	Prediction	Pattern finding
Example	Spam detection	Customer grouping

2. Classification Models

Classification = Data ko classes/categories me divide karna.

A) Decision Tree

Decision Tree ek tree structure use karta hai jo conditions ke basis par decisions leta hai.

Structure

- Root Node → Starting point
 - Decision Node → Condition check
 - Leaf Node → Final output
-

Working Algorithm

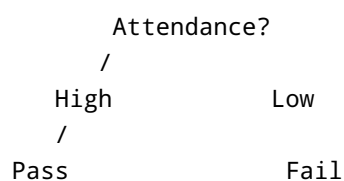
Decision Tree Algorithm Steps

1. Dataset lo
 2. Best feature select karo
 3. Data split karo
 4. Har subset par repeat karo
 5. Final leaf node prediction deta hai
-

Example Dataset

Attendance	Study Hours	Result
High	5	Pass
High	2	Pass
Low	1	Fail
Low	3	Fail

Decision Tree Representation



Solved Example

New Student:

- Attendance = High

Tree ke according:

High → Pass

Final Prediction = PASS

Important Concepts in Decision Tree

1. Entropy

Entropy data ki impurity/randomness measure karti hai.

- Low entropy = Data pure hai
- High entropy = Data mixed hai

Entropy Formula

$$Entropy(S) = -P_1 \log_2(P_1) - P_2 \log_2(P_2)$$

Where:

- P1 = Probability of Yes
 - P2 = Probability of No
-

Entropy Example

Suppose dataset:

Result
Pass
Pass
Fail
Fail

Total = 4

Pass = 2

Fail = 2

$$P(Pass) = 2/4 = 0.5$$

$$P(Fail) = 2/4 = 0.5$$

Apply formula:

$$\begin{aligned} Entropy(S) &= -(0.5) \log_2(0.5) - (0.5) \log_2(0.5) \\ &= -(0.5)(-1) - (0.5)(-1) \\ &= 0.5 + 0.5 = 1 \end{aligned}$$

Final Entropy = 1

Maximum impurity.

Pure Dataset Example

Result
Pass
Pass
Pass
Pass

$$Entropy = 0$$

Means dataset completely pure hai.

2. Information Gain

Information Gain batata hai ki kaunsa feature best split de raha hai.

Higher Information Gain = Better split.

Information Gain Formula

$$IG(S, A) = Entropy(S) - Weighted Entropy(A)$$

Decision Tree Algorithm using Entropy & Information Gain

1. Dataset lo
 2. Entropy calculate karo
 3. Har feature ka Information Gain nikalo
 4. Highest Information Gain wala feature choose karo
 5. Data split karo
 6. Recursively repeat karo
 7. Final leaf node prediction deta hai
-

Solved Example using Entropy & Information Gain

Dataset

Weather	Play Cricket
Sunny	No
Sunny	No
Rainy	Yes
Rainy	Yes

Step 1: Calculate Total Entropy

Total records = 4

Yes = 2

No = 2

$$P(Yes) = 2/4 = 0.5$$

$$P(No) = 2/4 = 0.5$$

Entropy:

$$\begin{aligned} Entropy(S) &= -(0.5) \log_2(0.5) - (0.5) \log_2(0.5) \\ &= 1 \end{aligned}$$

Step 2: Split on Weather

Sunny

Weather	Play
Sunny	No
Sunny	No

Entropy(Sunny):

All No

$$Entropy(Sunny) = 0$$

Rainy

Weather	Play
Rainy	Yes
Rainy	Yes

Entropy(Rainy):

All Yes

$$Entropy(Rainy) = 0$$

Step 3: Weighted Entropy

$$\text{Weighted Entropy} = \frac{2}{4}(0) + \frac{2}{4}(0) \\ = 0$$

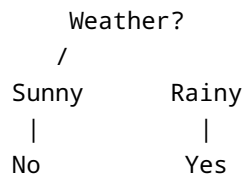
Step 4: Information Gain

$$IG = Entropy(S) - \text{Weighted Entropy}$$

$$IG = 1 - 0$$

$$IG = 1$$

Final Decision Tree



Final Prediction

- Sunny → Do Not Play
- Rainy → Play

Advantages

✓ Easy to understand ✓ Visualization simple ✓ Fast prediction

Disadvantages

✗ Overfitting issue ✗ Large tree complex ho jata hai

B) k-Nearest Neighbor (k-NN)

k-NN nearest neighbors dekhkar prediction karta hai.

k-NN Algorithm Steps

1. k choose karo
2. New point lo
3. Sab points se distance calculate karo
4. Nearest k neighbors choose karo
5. Majority class predict karo

Distance Formula

Euclidean Distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Solved Example

Training Data

Point	Class
(1,1)	Red
(2,2)	Red
(5,5)	Blue

New Point = (2,1)

k = 2

Step 1: Distance from (1,1)

$$d = \sqrt{(2 - 1)^2 + (1 - 1)^2}$$

$$d = \sqrt{1 + 0} = 1$$

Step 2: Distance from (2,2)

$$d = \sqrt{(2 - 2)^2 + (1 - 2)^2}$$

$$d = \sqrt{0 + 1} = 1$$

Step 3: Distance from (5,5)

$$d = \sqrt{(2 - 5)^2 + (1 - 5)^2}$$

$$d = \sqrt{9 + 16} = 5$$

Final Prediction

Nearest neighbors:

- Red
- Red

Output = RED

Advantages

✓ Simple ✓ No training phase ✓ Good for small datasets

Disadvantages

✗ Slow for large datasets ✗ Sensitive to noise ✗ Proper k value required

C) Naive Bayes

Naive Bayes probability-based classifier hai.

Ye Bayes Theorem use karta hai.

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Terms

Term	Meaning
P(A B)	P(B A)
P(B A)	P(A B)
P(A)	Probability of A
P(B)	Probability of B

Solved Example

Suppose:

- Total emails = 100
- Spam emails = 40

- Emails containing "WIN" = 30
- Spam emails containing "WIN" = 24

Find probability:

$$P(\text{Spam}|\text{WIN})$$

Step 1

$$P(\text{Spam}) = 40/100 = 0.4$$

Step 2

$$P(\text{WIN}|\text{Spam}) = 24/40 = 0.6$$

Step 3

$$P(\text{WIN}) = 30/100 = 0.3$$

Apply Formula

$$\begin{aligned} P(\text{Spam}|\text{WIN}) &= \frac{0.6 \times 0.4}{0.3} \\ &= 0.8 \end{aligned}$$

Final Answer

Probability that email is spam = 80%

Advantages

✓ Fast ✓ Good for text classification ✓ Works with large datasets

Disadvantages

✗ Assumes feature independence ✗ Complex relationships detect nahi karta

3. Clustering Algorithms

Clustering = Similar data points ko groups me divide karna.

Ye unsupervised learning ka part hai.

A) K-Means Clustering

K-Means data ko K clusters me divide karta hai.

K-Means Algorithm Steps

1. K choose karo
 2. Random centroids select karo
 3. Nearest centroid assign karo
 4. New centroid calculate karo
 5. Repeat until stable
-

Solved Example

Data Points

Point
2
4
10
12

K = 2

Step 1: Choose Centroids

Suppose:

- C1 = 2
 - C2 = 10
-

Step 2: Assign Clusters

Point	Distance from C1	Distance from C2	Cluster
2	0	8	C1
4	2	6	C1
10	8	0	C2
12	10	2	C2

Step 3: Update Centroids

Cluster C1:

$$(2 + 4)/2 = 3$$

Cluster C2:

$$(10 + 12)/2 = 11$$

New Centroids:

- C1 = 3
- C2 = 11

Final Clusters

Cluster 1 = {2,4}

Cluster 2 = {10,12}

Advantages

✓ Fast ✓ Easy ✓ Good for large datasets

Disadvantages

✗ K manually choose karna padta hai ✗ Outliers affect karte hain

B) Hierarchical Clustering

Tree-based clustering technique.

Types

1. Agglomerative

Bottom-up approach

2. Divisive

Top-down approach

Agglomerative Algorithm

1. Har point separate cluster
 2. Closest clusters merge karo
 3. Repeat until one cluster bache
-

Solved Example

Data Points:

A = 1 B = 2 C = 10

Step 1

Clusters:

{A} {B} {C}

Step 2

Nearest: A and B

Merge:

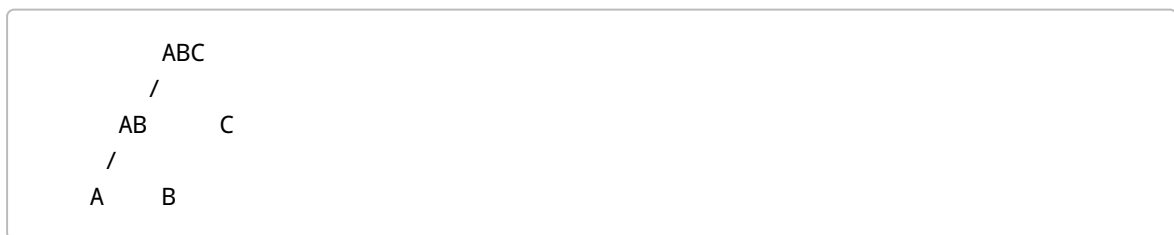
{AB} {C}

Step 3

Final Merge:

{ABC}

Dendrogram



Advantages

✓ Better visualization ✓ No need of K initially

Disadvantages

✗ Slow for large datasets ✗ Complex calculations

K-Means vs Hierarchical

Feature	K-Means	Hierarchical
Speed	Fast	Slow
K Required	Yes	No
Structure	Flat	Tree

4. Dimensionality Reduction

High-dimensional data = Too many features.

Problems:

- Slow training
- Complex computation
- Overfitting

Solution: Dimensionality Reduction

PCA (Principal Component Analysis)

PCA important information preserve karke dimensions reduce karta hai.

PCA Working

1. Data standardize karo
 2. Covariance matrix banao
 3. Eigenvalues/Eigenvectors find karo
 4. Principal components choose karo
 5. Data transform karo
-

Example

Original Features:

- Height
- Weight
- Income
- Spending
- Age

PCA:

5 features → 2 principal components

Principal Components

Component	Meaning
PC1	Maximum variance
PC2	Second maximum variance

Advantages

✓ Reduces complexity ✓ Faster training ✓ Better visualization

Disadvantages

✗ Information loss possible ✗ Hard interpretation

5. Association Rule Mining

Association Rule Mining relationships find karta hai.

Mostly shopping data analysis me use hota hai.

Example

"Bread lene wale Butter bhi lete hain."

Apriori Algorithm

Frequent itemsets find karta hai.

Apriori Algorithm Steps

1. Frequent items find karo
 2. Minimum support check karo
 3. Candidate itemsets generate karo
 4. Frequent itemsets select karo
 5. Rules generate karo
-

Important Terms

1. Support

$$Support(A) = \frac{Transactions\ containing\ A}{Total\ Transactions}$$

2. Confidence

$$Confidence(A \rightarrow B) = \frac{Support(A \cup B)}{Support(A)}$$

Solved Example

Transactions

Transaction	Items
T1	Bread, Butter
T2	Bread, Milk
T3	Bread, Butter
T4	Butter

Find Support of Bread

Bread appears in:

- T1
- T2
- T3

Total transactions = 4

$$Support(Bread) = 3/4 = 0.75$$

Find Confidence(Bread→Butter)

Bread and Butter together:

- T1
- T3

$$Confidence = 2/3 = 0.67$$

Final Answer

Support = 75%

Confidence = 67%

Applications

✔ Product recommendation ✔ Market basket analysis ✔ Cross-selling

6. Model Evaluation

Model ki performance evaluate karne ke liye metrics use karte hain.

A) Confusion Matrix

Actual / Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

Terms

Term	Meaning
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

Example

Suppose:

TP = 40 TN = 50 FP = 5 FN = 5

B) Accuracy

Overall correct predictions.

Formula

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Solved Example

$$\begin{aligned} Accuracy &= \frac{40 + 50}{40 + 50 + 5 + 5} \\ &= 90/100 = 0.9 \end{aligned}$$

Accuracy = 90%

C) Precision

Positive predictions me kitne correct hain.

Formula

$$Precision = \frac{TP}{TP + FP}$$

Solved Example

$$\begin{aligned} Precision &= \frac{40}{40 + 5} \\ &= 40/45 = 0.89 \end{aligned}$$

Precision = 89%

D) Recall

Actual positives me kitne detect hue.

Formula

$$Recall = \frac{TP}{TP + FN}$$

Solved Example

$$\begin{aligned} \text{Recall} &= \frac{40}{40 + 5} \\ &= 0.89 \end{aligned}$$

Recall = 89%

E) F1-Score

Precision aur Recall ka balance.

Formula

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Solved Example

Precision = 0.89 Recall = 0.89

$$\begin{aligned} F1 &= \frac{2 \times 0.89 \times 0.89}{0.89 + 0.89} \\ &= 0.89 \end{aligned}$$

F1 Score = 89%

Accuracy vs Precision vs Recall vs F1

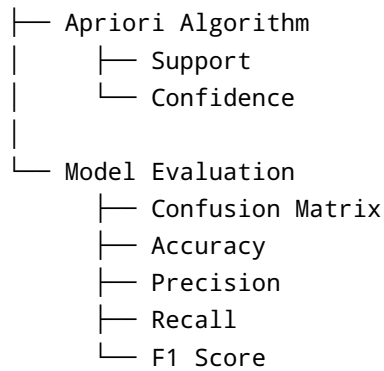
Metric	Meaning
Accuracy	Overall correctness
Precision	Correct positive predictions
Recall	Actual positives detected
F1 Score	Balance between precision & recall

Real-Life Importance

Metric	Important In
Accuracy	General prediction
Precision	Spam detection
Recall	Disease detection
F1 Score	Imbalanced datasets

Complete Quick Revision Mind Map

```
MACHINE LEARNING
|
├── Supervised Learning
│   ├── Classification
│   └── Regression
|
├── Unsupervised Learning
│   └── Clustering
|
├── Decision Tree
│   ├── Tree structure
│   └── Conditions
|
├── k-NN
│   ├── Distance
│   └── Neighbors
|
├── Naive Bayes
│   ├── Probability
│   └── Bayes theorem
|
├── K-Means
│   ├── Centroids
│   └── Clusters
|
├── Hierarchical Clustering
│   ├── Agglomerative
│   └── Divisive
|
└── PCA
    ├── Dimensionality reduction
    └── Principal components
```



Important Viva Questions

1. What is Machine Learning?

Machine Learning is a technique where computers learn patterns from data automatically.

2. Difference between supervised and unsupervised learning?

Supervised	Unsupervised
Labeled data	Unlabeled data
Prediction	Pattern finding

3. What is classification?

Classification means dividing data into categories/classes.

4. What is clustering?

Clustering means grouping similar data points together.

5. What is K in K-Means?

K represents number of clusters.

6. What is PCA?

PCA is a dimensionality reduction technique.

7. What is support in Apriori?

Support measures how frequently an item appears.

8. What is confusion matrix?

A table used to evaluate classification models.

9. Difference between precision and recall?

Precision	Recall
Correct positive predictions	Actual positives detected

10. What is F1 Score?

F1 Score is harmonic mean of precision and recall.